FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE JEE (Advanced), 2015

FULL TEST – V

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 234**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 – 08)** contains 8 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (09 – 12) contains 4 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

Section-A (13 to 18) contains 3 paragraphs with each having 2 questions. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-C (01 – 05)** contains 5 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

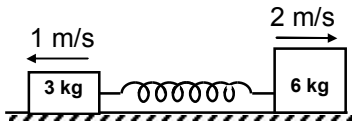
PART – I

SECTION – A

Straight Objective Type

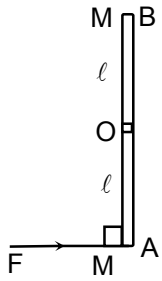
This section contains **8 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** option **is correct**.

1. Two blocks of mass 3 kg and 6 kg respectively are placed on a smooth horizontal surface. They are connected by a light spring of force constant k=200 N/m. Initially the spring is unstretched. The indicated velocities are imparted to the blocks. The maximum extension of the spring will be



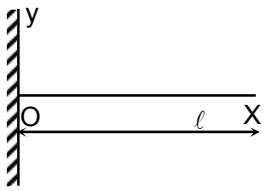
(A) 30 cm (B) 25 cm
(C) 20 cm (D) 15 cm

2. Two identical rods of mass M and length ℓ are hinged together as shown in the figure and lies at rest on smooth horizontal plane. A force F acts on end A of rod OA. The magnitude of instantaneous acceleration of hinge O is



(A) $\frac{F}{M}$ (B) $\frac{2F}{M}$
(C) $\frac{F}{2M}$ (D) None of these

3. Transverse wave are set in a thin rod clamped at one end O. The wave velocity is v. The rod vibrates in the 4th overtone mode with amplitude y_0. The equation of motion of the point at x = 2ℓ/3 is



(A) $y = y_o \sin\frac{5\pi}{3}\cos\frac{5\pi vt}{2\ell}$

(B) $y = y_o \sin\frac{7\pi}{3}\cos\frac{7\pi vt}{2\ell}$

(C) $y = y_0 \sin\frac{10\pi}{3}\cos\frac{9\pi vt}{2\ell}$

(D) y = 0

Space for Rough work

4. A rigid box contain one mole of monatomic ideal gas. The walls of the box have total surface area A, thickness b and thermal conductivity K. Initially the gas is at temperature T_0 and pressure P_0. The temperature of surroundings is ½ T_0

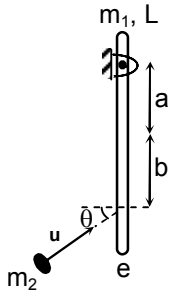
(A) The temperature as a function of time is $T = \frac{T_0}{4}\left[1+e^{-\frac{2KAt}{bR}}\right]$

(B) The temperature as a function of time is $T = \frac{T_0}{4}\left[1+e^{-\frac{2KAt}{3bR}}\right]$

(C) The pressure of gas as a function of time is $P = \frac{P_0}{4}\left[1+e^{-\frac{2KAt}{3bR}}\right]$

(D) None of these

5. A smooth uniform straight rod of mass m_1 and length L is placed on a smooth horizontal table and hinged at a distance 'a' from centre. A particle of mass m_2 moving horizontally with speed u hits the rod at angle θ from normal. Coefficient of restitution between rod and particle is e. The angular velocity of rod just after collision.



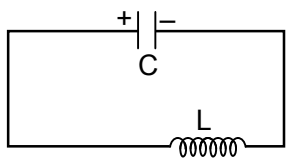
(A) $\frac{m_2(a+b)(e+1)u\cos\theta}{\left[m_1\left(\frac{L^2}{12}+a^2\right)+m_2(a+b)^2\right]}$

(B) $\frac{m_1(a+b)(e+1)u\cos\theta}{\left[m_2\left(\frac{L^2}{12}+a^2\right)+m_2(a+b)^2\right]}$

(C) $\omega = 0$

(D) None of these

6. In an LC circuit shown in the figure, C = 1F and L = 4H, at t = 0 charge on capacitor is 4C and it is decreasing at a rate of $\sqrt{5}$ C/sec. If maximum charge and current in the circuit are q_{max} and i_{max} then the value of i_{max}/q_{max} is



(A) $2s^{-1}$

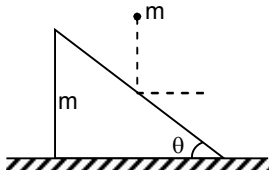
(B) $0.5\ s^{-1}$

(C) $\sqrt{\frac{2}{3}}\ s^{-1}$

(D) none of these

Space for Rough work

7. A particle of mass 'm' is dropped on a triangular wedge of same mass. Immediately after collision velocity of particle is in horizontal direction. If the surfaces are smooth. The maximum possible value of θ so that the above mentioned phenomenon takes place is



(A) $\theta < \tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (B) $\theta > \tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

(C) $\theta > 45°$ (D) $\theta < 30°$

8. In an R-L-C circuit v = 20 sin (314 t + $5\pi/6$) and i = 10 sin (314 t + $2\pi/3$). The power factor of the circuit is

(A) 0.5 (B) 0.966

(C) 0.866 (D) 1

Multiple Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

9. Bichromatic light of wavelength $\lambda_1 = 5000\,\text{Å}$ and $\lambda_2 = 7000\,\text{Å}$ are used in YDSE. Then

(A) 14th order maxima of λ_1 will coincide with 10th order maxima of λ_2

(B) 21st order maxima of λ_2 will coincide with 15th order maxima of λ_1

(C) 11th order minima of λ_1 will coincide with 8th order minima of λ_2

(D) 3rd order minima of λ_1 will coincide with 4th order minima of λ_2

10. P, Q and R are three particles of a medium which lie on the x-axis. A sine wave of wave length λ is traveling through the medium in the x-direction. P and Q always have the same speed, while P and R always have the same velocity. The minimum distance between

(A) P and Q is $\lambda/2$ (B) P and Q is λ

(C) P and R is $\lambda/2$ (D) P and R is λ

Space for Rough work

11. Which of the following statements is/are true?
(A) In vacuum the speed of red color is more than that of violet color
(B) An object in front of a mirror is moved towards the pole of a spherical mirror from infinity, it is found that image also moves towards the pole. The mirror must be convex.
(C) There exist two angles of incidence in a prism for which angles of deviation are same except minima deviation.
(D) A ray travels from a rarer medium to denser medium. There exist three angles of incidence for which the deviation is same.

12. Ten gm of oxygen at STP is expanded to a volume of 14×10^{-3} m^3. Then
(A) The final pressure is 0.5 atm in case of isothermal process
(B) The final pressure is $1/2^{7/5}$ atm in case of adiabatic process
(C) The final temperature is greater than 273K in case of adiabatic process
(D) The final temperature is 273K in case of isothermal process

Comprehension Type

This section contains **3 paragraphs**, each describing theory, experiments, data etc. Six questions relate to the three paragraphs with **two questions on each paragraph**. Each question has **only one** correct answer among the four given options (A), (B), (C) and (D).

Paragraph for Questions 13 & 14

When a surface is irradiated with light of wavelength 4950 $\mathring{A}$, a photocurrent appears which vanishes if a retarding potential greater than 0.6 volts is applied across the phototube. When a second source of light is used, it is found that the retarding potential is changed to 1.1 volt.

13. The work function of the emitting surface is:
(A) 2.2 eV (B) 1.5 eV
(C) 1.9 eV (D) 1.1 eV

14. The wavelength of the second source is:
(A) 6150 $\mathring{A}$ (B) 5150 $\mathring{A}$
(C) 4125 $\mathring{A}$ (D) 4500 $\mathring{A}$

Space for Rough work

Paragraph for Questions 15 & 16

The atomic masses of the hydrogen isotopes are:

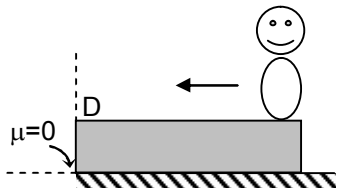
Hydrogen $_1H^1$ = 1.007825 amu
Deuterium $_1H^2$ = 2.014102 amu
Tritium $_1H^3$ = 3.016049 amu

15. The number of fusion reactions required to generate 1 kWh is nearly
(A) 10^8 (B) 10^{18}
(C) 10^{28} (D) 10^{38}

16. The mass of deuterium, $_1H^2$ that would be needed to generate 1 kWh
(A) 3.7 kg (B) 3.7 g
(C) 3.7×10^{-5} kg (D) 3.7×10^{-8} kg

Paragraph for Questions 17 & 18

A man of mass 50 kg is running on a plank of mass 150 kg with speed of 8 m/s relative to plank as shown in the figure. (both were initially at rest and the velocity of man with respect to ground any how remains constant). Plank is placed on smooth horizontal surface. The man, while running whistles with frequency f_0. A detector (D) is placed on plank detects frequency. The man jumps off with same velocity (with respect to ground) from point d and slides on the smooth horizontal surface [Assume coefficient of friction between man and horizontal is zero]. The speed of sound in still medium is 330 m/s. Answer following questions on the basis of above situations.



17. The frequency of sound detected by detector D, before man jumps off the plank is:
(A) $\frac{332}{324}f_0$ (B) $\frac{330}{322}f_0$
(C) $\frac{328}{336}f_0$ (D) $\frac{330}{338}f_0$

18. The frequency of sound detected by detector D, after man jumps off the plank is:
(A) $\frac{332}{324}f_0$ (B) $\frac{330}{322}f_0$
(C) $\frac{328}{336}f_0$ (D) $\frac{330}{338}f_0$

Space for Rough work

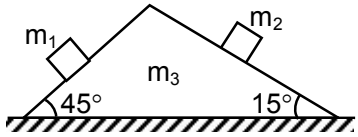
SECTION – C

Integer Answer Type

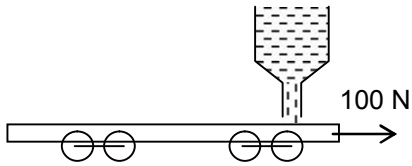
This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. To a man walking at the rate of 3 km/h the rain appears to fall vertically. When he increases his speed to 6 km/h it appears to meet him at an angle of 45° with vertical. The speed of rain is $n\sqrt{2}\,km/hr$. $\vec{V}_{RM} = \vec{V}_R - \vec{V}_M$ Find the value of n.

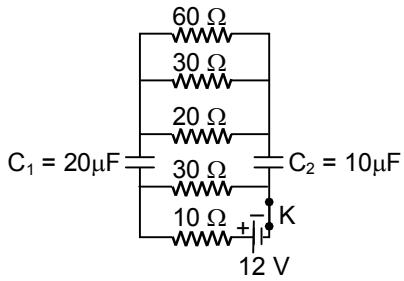
2. A wedge of mass m_3 is placed on a smooth horizontal surface. Blocks of mass m_1 and m_2 are placed on the smooth inclined faces of m_3. Angles of inclination are as shown. It is found that m_3 does not move. Find ratio of m_2/m_1.



3. A flat car of mass 50 kg starts moving to the right due to a constant force 100 N as shown in figure. S and spills on the flat car from a stationary hopper. The velocity of loading is constant and equal to 1 kg/s. The velocity of the car after 10 seconds. Is 100/n. Find n.



4. For the circuit arrangement shown in the figure, find the potential difference (in V) across C_2 in the steady state condition.



5. A charged particle is projected in a magnetic field $\vec{B}$ = $(3i + 4j) \times 10^{-2}$ T. The acceleration of the particle is found to be $\vec{a} = (-\frac{8}{3}\hat{i} + y\hat{j})$ m/s^2. Find the value of y.

Space for Rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

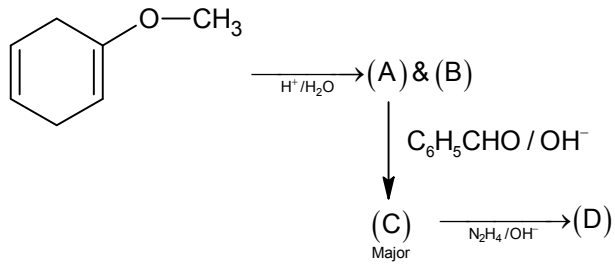
This section contains **8 multiple choice** questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONLY ONE** option is correct.

1. Which of the following is true for hydroxyl amine?
   (A) More basic than ammonia.
   (B) Disproportionates slowly in acidic solution producing N_2O and NH_4^+.
   (C) Disproportionates rapidly in acidic solution producing N_2 and NH_3.
   (D) Reacts with 2-methyl cyclohexanone to give cyclic amide of caproic acid.

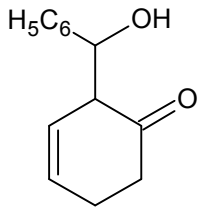
2. The molecule of P_4S_9 contains:
   (A) 4 $p\pi$ - $d\pi$ phosphorus – sulphur bonds.
   (B) P – S – P linkages.
   (C) Three phosphorus are tetrahedral unit and one phosphorus as pyramidal unit.
   (D) Has 4 P = S bonds.

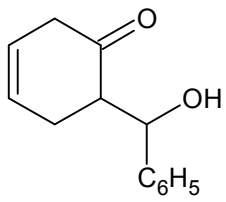
3. How many moles of HSCN will react with iodine, so that the produced I^- will react with neutral 10 ml of 0.2 M $KMnO_4$?
   (A) 0.16×10^{-4} (B) 1.6×10^{-4}
   (C) 1.2×10^{-4} (D) 0.12×10^{-5}

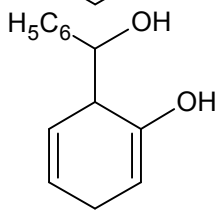
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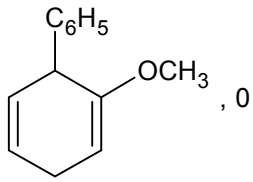
4. 

The structure of compound (C) and has many enantiomeric pairs are possible for (D)?

(A)  , 2

(B)  , 4

(C)  , 1

(D)  , 0

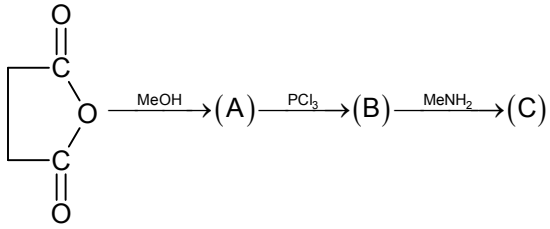
5. $CoCl_2$ gives a buff coloured precipitate with KCN, but with excess of KCN, the precipitate dissolves due to the formation of:
(A) $Co(CN)_2$ (B) $K_4[Co(CN)_6]$
(C) $K_3[Co(CN)_6]$ (D) $K_4[Co(CN)_4Cl_2$

6. The emf of the following cell is – 0.46 V.
$Pt(H_2),\ HSO_3^-(0.4\ M), SO_3^{2-}(6.4\times10^{-3}\ M)\,||\,Zn^{2+}(0.3M)\,|\,Zn$. If $E^o_{Zn^{2+}/Zn} = 0.76$ V, calculate pK_a of HSO_3^-, i.e. for equilibrium

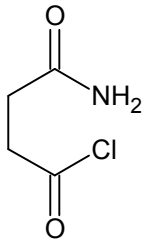
$HSO_3^- \rightleftharpoons H^+ + SO_3^{2-}$

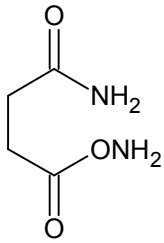
(A) 7.13 (B) 6.13
(C) 8 (D) 5.13

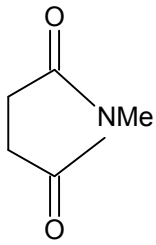
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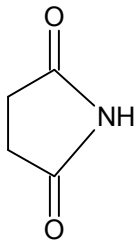
7. 

The product C is:

(A) 

(B) 

(C) 

(D) 

8. The ratio of masses of photons corresponding to transition with shortest wavelength of Lymann series of H-atom and lowest frequency of Paschen series of He^+ ion are:
(A) 7 : 36 (B) 5 : 9
(C) 9 : 5 (D) 36 : 7

Space for Rough work

Multiple Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

9. Which of the following statements are not correct?
   (1) From a mixed precipitate of AgCl and AgI, ammonia solution only dissolves AgCl.
   (2) $B(OH)_3 + NaOH \rightleftharpoons NaBO_2 + Na[B(OH)_4] + H_2O$ can be made to proceed forward by addition of trans-diol.
   (3) On passing H_2S gas in II group sometimes the solution turns milky indicating the presence of reducing agent.
   (4) $CuSO_4$ decolourises on addition of excess KCN, due to the formation of $[Cu(CN)_4]^{3-}$.

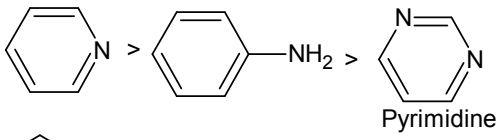
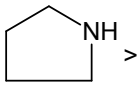
   (A) 1 is correct (B) 2 is incorrect
   (C) 3 is incorrect (D) 4 is correct

10. The reaction given below shows the enthalpy change for the chemical reactants between N_2, O_2, NO and NO_2 (taking the given data into account).

$$N_2(g) + O_2(g) \longrightarrow 2NO(g) \quad \Delta H = 110 \text{ kJ/mol}$$

$$2NO(g) + O_2(g) \longrightarrow 2NO_2(g) \quad \Delta H = -200 \text{ kJ/mol}$$

   Select the correct statements:
   (A) The standard heat of formation of NO_2 is 90 kJ mol^{-1}.
   (B) The oxidation reaction from N_2 to NO_2 is exothermic.
   (C) The standard heat of formation of NO is – 55 kJ mol^{-1}.
   (D) The oxidation reaction of NO to NO_2 is endothermic.

11. Which of the following is correct order of basic strength:
   (A) $BuNH_2 < Bu_2NH < Bu_3N$, order of basic strength having chlorobenzene as solvent.
   (B) $m-MeOC_6H_4NH_2 < o-MeOC_6H_4NH_3 < p-MeOC_6H_5NH_2$ order of basic strength in gaseous phase.
   (C)  Pyrimidine
   (D)  Pyrrolidine

12. Which of the following are correct statements:
   (A) Entropy of system increases when an ideal gas is expanded isothermally.
   (B) Gas with higher γ has high magnitude of slope in a P(y-axis) v/s V(x – axis) curve.
   (C) Final temperature in reversible adiabatic expansion is lesser than in irreversible adiabatic expansion.
   (D) Vapourisation at boiling point, $\Delta G < 0$.

Space for Rough work

Comprehension Type

This section contains **3 paragraphs** each describing theory, experiment, data etc. **Six questions** relate to three paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct** answers among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 13 to 14

Chlorine on reaction with compound (X) in basic medium, according to the given equation:

$IO_3^- + xOH^- + Cl_2 \longrightarrow (A) + yH_2O + zCl^-$

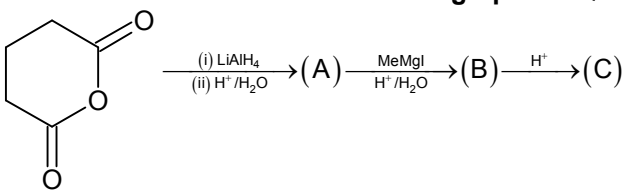
Compound (A) on heating in acidic medium $(A) \xrightarrow[-4H_2O]{100^\circ C} (B) \xrightarrow{200^\circ C} (C)$

13. Identify the compound (C)?
    (A) I_2O_3 (B) I_2
    (C) I_2O_7 (D) I_2O_5

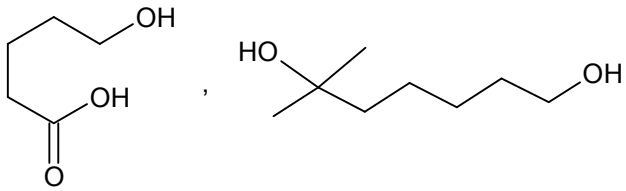
14. The stoichiometric coefficient in the above (x), (y), (z) respectively are:
    (A) 6, 3, 2 (B) 5, 4, 1
    (C) 3, 4, 2 (D) 6, 2, 3

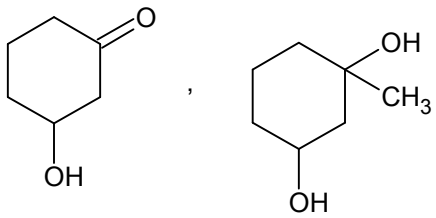
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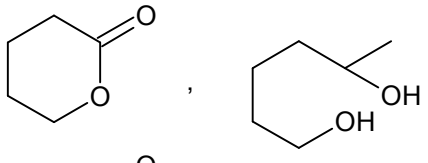
Paragraph for Question Nos. 15 to 16

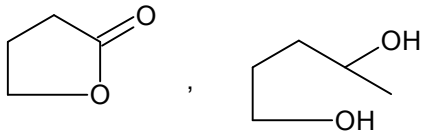


15. The structure of compound (A) and (B) respectively are:

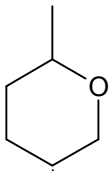
(A) 

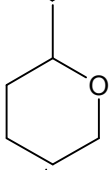
(B) 

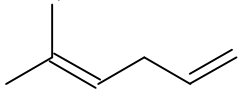
(C) 

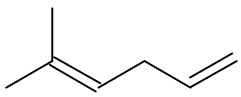
(D) 

16. Which of the following statement is correct for compound (C)?

(A) (C) is  and is optically active.

(B) (C) is  and is optically inactive.

(C) (C) is  shows geometrical isomerism.

(D) (C) is  shows no geometrical isomerism.

Space for Rough work

Paragraph for Question Nos. 17 to 18

The vapour density of equilibrium mixture for
$N_2O_4(g) \rightleftharpoons 2NO_2(g)$ is 28.75 at 1 atm.
This mixture is allowed to diffuse out pressure for 10 seconds.

17. Calculate the respective mole fraction of the two gases in N_2O_4 and NO_2 in the effusing out mixture?
(A) $X_{N_2O_4} = 0.188, \quad X_{NO_2} = 0.812$
(B) $X_{N_2O_4} = 0.812, \quad X_{N_2O_4} = 0.188$
(C) $X_{N_2O_4} = 0.212, \quad X_{NO_2} = 0.788$
(D) $X_{NO_2} = 0.788, \quad X_{NO_2} = 0.212$

18. Calculate new K_P of reaction?
(A) 1.875
(B) 2.2
(C) 1.42
(D) 2.8

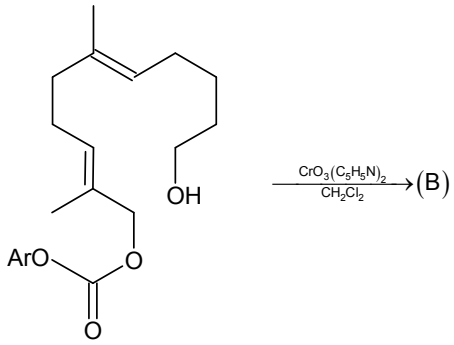
Space for Rough work

SECTION – C

Integer Answer Type

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. How many halogen atoms are bridging in $As_2Cl_8^{2-}$?

2. 

The number of stereoisomer possible for the compound (B) are:

3. The total number of tetrahedral units in the monomers of Buna-N?

4. While studying the decomposition of gaseous compound it is observed that a plot of logarithm of its partial pressure versus time is linear. What is the order of decomposition?

5. In the sodium chloride structure each Na^+ is surrounded by $\frac{6x}{2}$ next nearest Na^+ ions. What is the value of x?

Space for Rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **8 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** option **is correct**.

1. If $[x^3 + x^2 + x + 1] = [x^3 + x^2 + 1] + x$, (where [.] denote the greatest integer function), then number of solutions of the equation $\log |[x]| = 2 - |[x]|$ is
(A) 1 (B) 0
(C) 3 (D) 2

2. A is a set containing 7 elements and a subset A_1 of A is chosen. Then set A is reconstructed by replacing the elements of A_1. Now another subset A_2 of A is chosen. In this way 4 subsets A_1, A_2, A_3, A_4 are formed. The number of ways of choosing A_1, A_2, A_3, A_4 so that these subsets are pair wise disjoint, is
(A) 5^7 (B) 7^5
(C) 4^7 (D) 840

3. Let C_0, C_1, C_2 ....., C_n be the binomial co-efficient in the expansion of $(1 + x)^n$, $n \in N$.
If $S_n = \frac{2C_0}{2} + \frac{2^2C_1}{6} + \frac{2^3C_2}{12} ..... \frac{2^{n+1}C_n}{(n+1)(n+2)} + \frac{1}{n+1} + \frac{1}{2(n+1)(n+2)}$, then the value of $\frac{S_7}{S_6}$ is
(A) 7 (B) $\frac{7}{6}$
(C) $\frac{7}{3}$ (D) $\frac{7}{2}$

Space for rough work

4. The number of complex numbers satisfying the conditions $\left|\frac{z}{\bar{z}}+\frac{\bar{z}}{z}\right|=1$, |z| = 1 and arg (z) $\in (0, 2\pi)$ is
(A) 1 (B) 2
(C) 4 (D) 8

5. $\frac{1}{2}\tan\frac{\theta}{2}+\frac{1}{4}\tan\frac{\theta}{4}$ is equal to
(A) $4\cot\frac{\theta}{4}-\cot\theta$ (B) $\cot\frac{\theta}{4}-\cot\theta$
(C) $\frac{1}{4}\cot\frac{\theta}{4}-\cot\theta$ (D) $\cot\frac{\theta}{4}-4\cot\theta$

6. Equation of the parabola with latus rectum $2\sqrt{2}$ units and axis as the line x – y = 0 and tangent at the vertex as x + y = 1, is (origin lies outside parabola)
(A) $x^2 + y^2 – 2xy + 4x + 4y + 4 = 0$ (B) $x^2 + y^2 – 2xy + 8x + 8y – 8 = 0$
(C) $x^2 + y^2 – 2xy – 4x – 4y + 4 = 0$ (D) $x^2 + y^2 – 2xy – 2x – 2y + 2 = 0$

7. If the circles $x^2 + y^2 + (3 + \sin\beta)x + (2\cos\alpha)y = 0$ and $x^2 + y^2 + (2\cos\alpha)x + 2cy = 0$ touch each other then the maximum value of 'c' is
(A) $\frac{1}{2}$ (B) 1
(C) $\frac{3}{2}$ (D) 2

8. If f be a continuous function on [0, 1], differentiable in (0, 1) such that f(1) = 0, then exists some c $\in$ (0, 1] such that
(A) cf′(c) – f(c) = 0 (B) f′(c) + cf(c) = 0
(C) f′(c) – cf(c) = 0 (D) cf′(c) + f(c) = 0

Space for rough work

Multiple Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct

9. Let $f(x) = \begin{cases} 5x+1 & \text{if} \quad x \le 2 \\ \int_0^x (5+|1-t|)\,dt & \text{if} \quad x > 2 \end{cases}$

then
(A) f(x) is continuous but not differentiable at x = 2
(B) f(x) is not continuous at x = 2
(C) f(x) is differentiable everywhere, $\forall\ x \in (2, \infty)$
(D) the right hand derivative of f(x) at x = 3 does not exist

10. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, $\vec{a} \times \vec{c} = \vec{b} + \lambda\vec{a}$ and $\vec{a} \cdot \vec{c} = 1$ then which of following is true

(A) $[\vec{a}\ \vec{b}\ \vec{c}] = -\frac{8}{3}$ (B) $\lambda = -\frac{2}{3}$

(C) $[\vec{a}\ \vec{b}\ \vec{c}] = -\frac{7}{3}$ (D) $\lambda = -\frac{1}{3}$

11. A right angle triangle ABC, right angle at A is inscribed in hyperbola $xy = c^2$ $(c > 0)$ such that slope of BC is 2. If distance of point A from centre of $xy = c^2$ is $\sqrt{10}$, then which of the following is/are correct for $xy = c^2$
(A) the value of c is 2
(B) the value of c is 4
(C) the equation of normal at point A can be $y = 2x - 3\sqrt{2}$
(D) the equation of normal at point A can be $y = 3x + 8\sqrt{2}$

12. For the equation $|x^2 - a| = \cos^{-1}\cos x$ to be inconsistent, the value of 'a' can be

(A) $\frac{1}{2}$ (B) $\frac{1}{6}$

(C) $-\frac{1}{2}$ (D) $-\frac{2}{3}$

Space for rough work

Comprehension Type

This section contains **3 paragraphs** each describing theory, experiment, data etc. **Six questions** relate to three paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct** answers among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 13 to 14

Read the following write up carefully and answer the following questions:

Consider three points A, B and C lying on $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (a > b) such that origin is a point interior to the ΔABC. Lines through A, B, C are drawn perpendicular to major axis to cut circle $x^2 + y^2 = a^2$ at P, Q, R respectively such that P, Q, R are on same side w.r.t major axis as that of A, B, C respectively

13. If ΔPQR is equilateral then area of ΔABC is

(A) $\frac{\sqrt{3}}{4}ab$ (B) $\frac{3\sqrt{3}}{4}ab$

(C) $\frac{3\sqrt{3}}{2}ab$ (D) none of these

14. Let ΔPQR is not necessarily equilateral, A and B lies in first quadrant such that slopes of OA and OB are m_1 and m_2 respectively, where O is the origin. If the area of sector OAB of ellipse is given by $\frac{1}{2}ab\theta$ then θ is equal to

(A) $\tan^{-1}\left|\frac{ab(m_1-m_2)}{b^2+a^2m_1m_2}\right|$ (B) $\tan^{-1}\left|\frac{m_1-m_2}{1+m_1m_2}\right|$

(C) $\tan^{-1}\left|\frac{ab(m_1-m_2)}{a^2+b^2m_1m_2}\right|$ (D) $\tan^{-1}\left|\frac{ab(m_1+m_2)}{a^2-b^2m_1m_2}\right|$

Space for rough work

Paragraph for Question Nos. 15 to 16

Read the following write up carefully and answer the following questions:

Consider a ΔABC with AB = c, BC = a and CA = b. Internal and external bisectors of $\angle C$ intersect AB and AB produced respectively at L and M, such that CL = CM. AD is a diameter of circum-circle of ΔABC. The radius of circum-circle is R

15. The value of $a^2 + b^2$ is
(A) R^2 (B) $2R^2$
(C) $3R^2$ (D) $4R^2$

16. Which of the following is incorrect?
(A) DC = a (B) $\angle BAC = \angle CAD$
(C) $\angle ADC = \angle CBM$ (D) $\angle BAC = \frac{\pi}{4} + \frac{\theta}{2}$, where $\theta = \angle ACB$

Paragraph for Question Nos. 17 to 18

Read the following write up carefully and answer the following questions:

If the expression for the nth term of the infinite sequence 1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, ... is $\left[\sqrt{\alpha n} + \frac{1}{\beta}\right]$,

(where [.] denotes the greatest integer function) then

17. Let $a = \alpha$, $b = \alpha + 1$ and $c = \alpha + \beta + 1$ then number of numbers out of first 1000 natural numbers which are divisible by a, b or c is
(A) 764 (B) 867
(C) 734 (D) none of these

18. Let $a = \alpha$, $b = \alpha + 1$, $c = \alpha + \beta + 1$ and $d = 3\beta + 1$. The number of divisors of the number $a^c \cdot b^c \cdot c^b \cdot d^b$ which are of the form $4n + 1$, $n \in N$ is equal to
(A) 24 (B) 48
(C) 96 (D) none of these

Space for rough work

SECTION – C

Integer Answer Type

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. Let f : R $\rightarrow$ R defined as $f(x) = x^3 + x + 1$, $1 \le x \le 2$. The graph of y = g(x) is the reflection of graph of y = f(x) through the line y = x. If the domain of g(x) is [a, b], then |a – b| is _____

2. In a paper of mathematics there are 5 questions, such that the sum of the marks is 30 and the marks for any question is not less than 2 and not more than 8 (only integer marks are awarded). If the number of ways in which the marks can be awarded is ABC, where A, B, C $\in$ N and A, B, C $\le$ 9. Then value of A + B – C is _____

3. If f: [0, 2] $\rightarrow$ R, $f(x) = \sqrt{x^3 + 2 - 2\sqrt{x^3+1}} + \sqrt{x^3 + 10 - 6\sqrt{x^3+1}}$ then $\int_0^2 f(x)\,dx$ is equal to _____

4. The plane $\frac{x}{1} + \frac{y}{2} + \frac{z}{3} = 1$ intersect x–axis, y–axis, z–axis at A, B, C respectively. If the distance between origin and orthocentre of ΔABC is equal to k then the value of 7k is equal to _____

5. The value of $\left| \lim_{x \to \frac{3\pi}{4}} \left(\frac{4\sin^2 x \cos x - \cos x + \sin x}{\sin x + \cos x} \right) \right|$ is equal to _____

Space for rough work